Software Requirements Specification

for

Professor Rating Based on Course Outcome Feedback System

Version <1.2>

Prepared by

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Contents

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft 1	Sreeja Rao Surabhi Srujana Yampati	Initial draft of the SRS document (Introduction and Overall Descriptions)	11/03/26
Draft 2	P. Rishith G. Vijay	Specification of Requirements (functional and nonfunctional).	13/3/26
Draft 3	P. Rishith G. Vijay S. Koushik	Addition of Uml diagrams and finalization of the SRS Document	16/3/26

1 Introduction

The Professor Rating Based on Course Outcome Feedback System is a web-based platform developed to collect structured student feedback about professors in the university. The system enables students to anonymously evaluate professors and provide insights on how well the Course Outcomes (COs) of a course were taught and understood.

In many academic institutions, course feedback is collected manually or through basic survey forms that often lack structure and analytics. This system aims to improve the feedback process by introducing a centralized platform where feedback can be securely collected, validated using attendance constraints, and analysed for improving teaching effectiveness.

1.1 Document Purpose

This document describes the functional and non-functional requirements for the Professor Rating based on Course Outcome Feedback System. It defines the behavior of the platform, system interfaces, constraints, and performance expectations.

The goal of this document is to provide developers, project managers, instructors, and stakeholders with a clear understanding of the system requirements so that the system can be designed and implemented efficiently.

1.2 Product Scope

The Professor Rating based on Course Outcome Feedback System is a full-stack web application that enables students to anonymously submit ratings and feedback for professors and courses they are enrolled in. The system also allows students to evaluate specific Course Outcomes (CO1-CO5) to determine whether the intended learning objectives were properly covered and understood during the course. The platform includes an attendance validation mechanism, where students are allowed to submit feedback only if their attendance in a course is greater than or equal to 75%.

The primary goals of the system include:

• Providing a secure and anonymous feedback platform for students
• Enabling structured evaluation based on course outcomes
• Improving teaching quality through actionable insights
• Ensuring fairness by enforcing attendance eligibility rules

The platform will be developed using React for the frontend, Node.js with Express for backend APIs, and MongoDB for data storage.

1.3 Intended Audience and Document Overview

This document is intended for:

• Developers - to implement system functionality and architecture

• Instructor / Evaluators - to evaluate project design

• Testers - to verify that the system behaves according to requirements

• End Users(students) - to understand the features of the feedback platform

Document Structure:

Section	Description
Section 1	Introduction and background
Section 2	Overall system description
Section 3	Detailed functional requirements
Section 4	Non-functional requirements
Section 5	Additional system requirements

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
API	Application Programming Interface used for communication between frontend and backend.
CO	Course Outcome - a measurable learning objective defined for a course.
CRUD	Create, Read, Update, Delete operations performed on database records
DB	Database used to store system information
Mongo DB	NoSQL database used for storing application data
Node.js	JavaScript runtime used to implement backend services
React	JavaScript framework used for building the frontend interface
REST API	API architecture used for communication between client and server
SRS	Software Requirements Specification
UI	User Interface used by students to interact with the system

1.5 Document Conventions.

This document follows the IEEE Software Requirements Specification (SRS) standard to ensure clarity, consistency, and ease of understanding for all stakeholders involved in the Professor Rating based on Course Outcome Feedback System.

1.5.1.1 Formatting Conventions

• The document uses Arial font, size 12 for all text content.
• Section headings and subsection headings follow the hierarchical numbering format specified in the IEEE SRS template (e.g., 1, 1.1, 1.2, etc.).
• All text in the document is single-spaced with 1-inch margins on all sides.
• Tables are used where appropriate to organize definitions, requirements, and data elements clearly.

1.5.1.2 Naming Conventions

• Functional requirements are identified using the format F1, F2, F3, ….
• Use cases are labeled using the format U1, U2, U3, ….
• System components, modules, and actors are named using clear and descriptive identifiers to avoid ambiguity.

1.5.1.3 Requirement Conventions

• The keyword "shall" is used to indicate mandatory system requirements.
• The keyword "may" is used to describe optional functionality.

These conventions help maintain consistency and ensure that all stakeholders interpret the system requirements accurately throughout the development lifecycle.

1.6 References and Acknowledgments

References used for the development of this system:

- IEEE Software Requirements Specification Standard
- React.js Documentation
- Node.js and Express Documentation
- MongoDB Documentation
- Software Engineering Course Materials

2 Overall Description

2.1 Product Overview

The Professor Rating based on Course Outcome Feedback System is a web application designed to collect structured student feedback for academic evaluation purposes.

Currently, feedback collection processes in many institutions are either manual or implemented using basic survey tools that do not enforce eligibility conditions or maintain anonymity. This system addresses these limitations by providing a dedicated feedback platform integrated with student attendance records.

A key feature of the platform is the attendance-based eligibility validation, where students can submit feedback only if they have at least 75% attendance in the respective course. This ensures that feedback is provided by students who have actively participated in the course.

The system follows a three-tier architecture consisting of a front-end interface, backend application services, and a database layer. Students access the platform through a web interface where they can view enrolled courses and submit feedback forms. The backend system processes requests, validates eligibility, and stores feedback securely in the database. Overall, the platform acts as an intermediary between students, professors, and academic administration, enabling institutions to collect meaningful insights about teaching quality and course effectiveness.

Explanation of Components

• Students (Users) - Students who access the platform to submit anonymous feedback for courses and professors.
• Professors - Faculty members whose teaching performance and course delivery are evaluated through student feedback.
• Professor Feedback Platform - The main system responsible for managing feedback forms, eligibility validation, and data processing.
• Attendance Validation Module - Ensures that only students with ≥75% attendance can submit feedback.
• Backend API Services - Server-side services that handle request processing, validation, and communication with the database.
• Database (MongoDB) - Stores student information, course details, attendance data, and feedback records.
• Web Interface (Frontend) - A browser-based interface that allows students to interact with the system and submit feedback.

2.2 Product Functionality

The Professor Rating and Course Outcome Feedback Website provides a centralized platform that enables students to submit structured and anonymous feedback about professors and course delivery. The system allows students to evaluate teaching quality, provide feedback on course outcomes, and ensure that feedback is collected only from eligible students based on attendance requirements. The following list summarizes the major functions of the system at a high level.

• **Student Authentication and Login**
The system allows students to securely log in using institutional credentials to access the feedback platform.

• **Course Retrieval and Display**
The system retrieves and displays the list of courses that the student is enrolled in along with the corresponding professors.

• **Attendance-Based Eligibility Verification**
The system verifies the student's attendance for a selected course and allows feedback submission only if the attendance is greater than or equal to 75%.

• **Professor Rating System**
Students can rate professors based on predefined evaluation criteria such as teaching clarity, course organization, and communication effectiveness.

• **Course Outcome Evaluation**
Students can provide structured feedback on Course Outcomes (CO1-CO5) to indicate how well each learning objective was covered during the course.

• **Feedback Form Submission**
The system allows students to submit feedback forms after completing all required sections of the evaluation form.

• **Anonymous Feedback Storage**
All feedback submitted by students is stored anonymously to ensure privacy and encourage honest responses.

• **Duplicate Submission Prevention**
The system prevents students from submitting feedback more than once for the same course.

• **Feedback Data Storage and Management**
The system securely stores feedback data in the database for future analysis and academic evaluation.

• **Administrative Monitoring and Reporting**
Authorized administrators can access aggregated feedback reports and statistics to evaluate teaching performance and course effectiveness.

2.3 Design and Implementation Constraints

The following constraints affect the design and development of the system:

• **Attendance Requirement**
Feedback can be submitted only by students who have attendance greater than or equal to 75% in the respective course.

• **Anonymous Feedback**
Student feedback must remain completely anonymous, and the system must ensure that the identity of the student cannot be linked to the submitted feedback.

• **Technology Stack Constraint**
The system must be developed using the prescribed technology stack, which includes React for the frontend, Node.js with Express for the backend, and MongoDB as the database.

• **Single Submission Rule**
Once a student submits a feedback form for a course, the system must prevent them from submitting another feedback for the same course.

• **Mandatory Form Completion**
Students must complete all sections of the feedback form before the system allows submission.

• **Course Enrollment Restriction**
Students may submit feedback only for professors whose courses they are officially enrolled in.

2.4 Assumptions and Dependencies

The following assumptions and dependencies are considered during the design and development of the system:

Assumptions:
• **Availability of Student and Professor Data**
It is assumed that the university will provide access to existing student and professor databases required for the system.
• **Attendance Records Availability**
Student attendance data will be available and accessible to the system for validating feedback eligibility.
• **Access to Academic Records**
Relevant academic data such as student marks or course information will be available if required for system validation or reporting.

• **Internet and Browser Access**
Users are assumed to have access to a modern web browser and a stable internet connection to interact with the system.

Dependencies:
• The system depends on institutional databases for retrieving student, professor, course, and attendance information.
• The system depends on MongoDB database services for storing feedback data and system records.
• The system depends on Node.js runtime and server infrastructure for backend processing and API services.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The Feedback System provides web-based user interfaces that allow different types of users to interact with the platform. These interfaces enable students, professors, and administrators to access system functionalities according to their roles.

Student Interface

• *Login Page* - Allows students to securely log in using their institutional credentials. The system verifies the credentials and grants access to the feedback platform.
• *Student Dashboard* - Displays the list of courses in which the student is currently enrolled. The dashboard provides quick access to feedback forms for each course and displays the status of feedback submission.
• *Course Feedback Form* - Provides a structured form where students can rate professors and evaluate Course Outcomes (CO1-CO5). The form includes rating fields, text input for comments, and submission controls.
• *Attendance Eligibility Notification* - Displays the student's attendance status for the selected course. The system informs the student whether they meet the minimum attendance requirement (≥75%) needed to submit feedback.
• *Submission Confirmation Page* - After successful feedback submission, the system displays a confirmation message informing the student that their feedback has been recorded anonymously.

Professor Interface

• *Professor Dashboard* - Displays the courses taught by the professor and provides access to feedback summaries related to those courses.
• *Feedback Report Page* - Allows professors to view aggregated feedback results collected from students. Individual student identities are not shown to ensure anonymity.
• *Course Outcome Evaluation Summary* - Displays statistical summaries of how students rated the coverage and understanding of Course Outcomes (CO1-CO5) for each course.

Administrator Interface

• *Admin Dashboard* - Provides an overview of system activity, including the number of feedback submissions, courses evaluated, and system usage statistics.
• *Course Management Page* - Allows administrators to add, update, or remove course information stored in the system.
• *Professor Management Page* - Enables administrators to manage professor records and assign professors to courses.
• *Feedback Monitoring Page* - Allows administrators to review aggregated feedback reports and monitor the overall performance of courses and professors.
• *System Maintenance Controls* - Allows administrators to perform administrative tasks such as updating system configurations, managing data integrity and ensuring the proper functioning of the platform.

3.1.2 Hardware Interfaces

The Feedback System interacts with standard computing devices and server infrastructure required to run the web-based application. The system does not require specialized hardware components, but it relies on common devices used to access and host the platform.

Student Devices

• *Personal Computers and Laptops* -Students may access the system using desktop computers or laptops. These devices provide the display, keyboard, and input mechanisms required to navigate the platform, complete feedback forms, and submit evaluations.
• *Mobile Devices and Tablets* -The system supports access through smartphones and tablets using modern web browsers. Touchscreen interfaces allow students to interact with the platform, select rating values, and submit feedback forms.

Server Hardware

• *Application Server* - The backend application runs on a server responsible for executing the Node.js and Express backend services. This server processes user requests, validates attendance eligibility, handles authentication, and manages communication between the frontend interface and the database.
• *Database Server* - The MongoDB database operates on a database server responsible for storing and retrieving system data including student records, course information, attendance data, professor details, and submitted feedback.

Network Infrastructure

• *Internet Connectivity* - The system requires reliable internet connectivity to allow communication between user devices, application servers, and database servers.
• *Networking Equipment* - Standard networking hardware such as routers, switches, and institutional network infrastructure supports the transmission of data between system components.

3.1.3 Software Interfaces

The Feedback System interacts with several software components that support authentication, data processing, and storage. These software interfaces enable communication between the frontend interface, backend services, and institutional data systems.

• *Frontend Application (React)* - The system provides a web-based user interface built using React. This interface allows students, professors, and administrators to interact with the platform, submit feedback, and view reports.

• *Backend Services (Node.js and Express)* - The backend application provides REST API services that process requests from the frontend, enforce system rules such as attendance validation and duplicate feedback prevention, and manage system operations.

• *Database System (MongoDB)* - The system uses MongoDB to store application data including student records, course information, professor details, attendance data, and feedback submissions.

• *Institutional Academic Systems* - The platform may interact with existing university systems to retrieve student enrollment data and attendance records required for feedback eligibility verification.

3.2 Functional Requirements

The functional requirements define the core behaviors and services that the Professor Rating Based on Course Outcome Feedback System must provide. These requirements describe the actions the system must perform in response to user interactions and system events.

Student Related Requirements

F1. Student Authentication - The system shall allow students to securely log in using their institutional credentials.

F2. Course Retrieval - The system shall retrieve and display the list of courses in which the authenticated student is enrolled.

F3. Attendance Eligibility Verification - The system shall verify the student's attendance for a selected course and allow feedback submission only if the attendance is greater than or equal to 75%.

F4. Feedback Form Access - The system shall allow eligible students to access the course feedback form for the selected course.

F5. Professor Rating Submission - The system shall allow students to rate professors using predefined evaluation criteria.

F6. Course Outcome Evaluation - The system shall allow students to evaluate Course Outcomes (CO1–CO5) using structured rating inputs.

F7. Feedback Comment Submission - The system shall allow students to submit additional textual comments related to the course or professor.

F8. Anonymous Feedback Handling - The system shall store student feedback anonymously without linking responses to the student's identity.

F9. Duplicate Feedback Prevention - The system shall prevent students from submitting feedback more than once for the same course.

F10. Feedback Submission Confirmation - The system shall display a confirmation message after successful feedback submission.

Professor Related Requirements

F11. Professor Course Access - The system shall allow professors to view the list of courses assigned to them.

F12. Aggregated Feedback Viewing - The system shall allow professors to view aggregated feedback results for their courses.

F13. Course Outcome Feedback Summary - The system shall display statistical summaries of Course Outcome evaluations (CO1–CO5) for each course.

Administrator Related Requirements

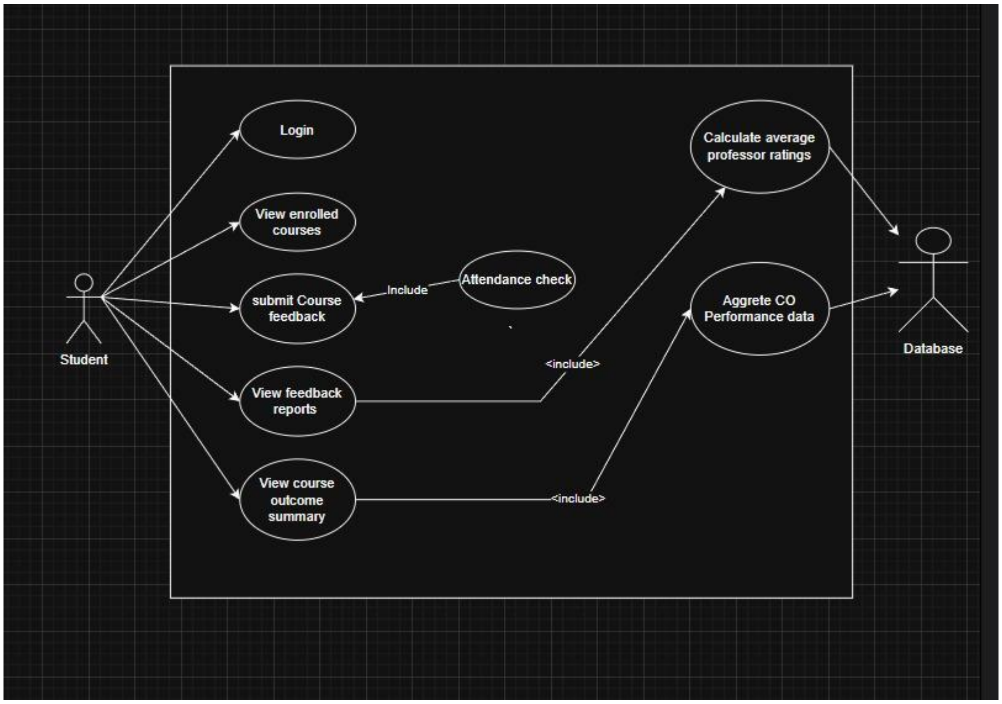
F14. Course Management - The system shall allow administrators to add, update, or remove course information.

F15. Professor Management - The system shall allow administrators to manage professor records and assign professors to courses.

F16. Feedback Monitoring - The system shall allow administrators to view aggregated feedback reports and system usage statistics.

3.3 Use Case Model

3.3.1 Use Case #1



Author – G. Vijay

Purpose - This use case diagram describes how students interact with the Professor Rating Based on Course Outcome Feedback System to submit course feedback, view reports, and access course outcome summaries. The system validates student eligibility through an attendance check before allowing feedback submission

Requirements Traceability - F1, F2, F3, F4, F5, F6, F7, F8, F9, F10

Priority - High

Preconditions –

- The student must be registered in the university system.
- The student must have valid login credentials.
- Course enrollment and professor information must exist in the system database
- Attendance data must be available for validation.

Post conditions –

- Student feedback is successfully submitted and stored anonymously.
- The system generates updated feedback reports and course outcome summaries.
- Dashboard analytics are updated with the latest feedback information.

Actors –

- Student
- Database

Flow of Events

1. Basic Flow

1. The student accesses the feedback platform.
2. The system displays the login interface.
3. The student enters login credentials.
4. The system authenticates the student and displays the student dashboard.
5. The student views the list of enrolled courses.
6. The student selects a course and chooses to submit course feedback.
7. The system performs an attendance check for the selected course.
8. If the attendance requirement (≥75%) is satisfied, the system displays the feedback form.
9. The student fills in professor ratings, course outcome evaluations, and optional comments.
10. The student submits the feedback form.
11. The system stores the feedback anonymously in the database.
12. The system processes feedback data to calculate average professor ratings.
13. The system aggregates Course Outcome performance data.
14. The system prepares dashboard analytics data for reporting.
15. The student may view feedback reports and course outcome summaries.

2. Alternative Flow

1. If the student attempts to submit feedback more than once for the same course, the system prevents duplicate submission.
2. The student may choose to only view feedback reports or course outcome summaries without submitting new feedback.

3. Exceptions

1. If attendance is below the required threshold (75%), the system prevents feedback submission.
2. If the system cannot retrieve attendance data, the feedback process is temporarily blocked.
3. Database errors may prevent feedback storage.

Includes –

- Attendance Check
- Calculate Average Professor Ratings
- Aggregate CO Performance Data
- Prepare Dashboard Analytics Data

Notes/Issues - Student identities must not be directly linked to feedback responses to ensure anonymity.

3.3.2 Use Case #2

Author – S. Koushik

Purpose - This use case describes how the administrator interacts with the Professor Rating Based on Course Outcome Feedback System to manage academic data such as courses and professors, monitor feedback reports, and generate analytical reports for evaluations.

Requirements Traceability - F14, F15, F16

Priority – Medium

Preconditions –

- The administrator must be registered in the system.
- The administrator must log into the system using valid credentials.
- Course, professor, and feedback data must exist in the system database.

Post conditions –

- Course and professor information may be added, updated, or removed.
- Feedback reports are generated and available for administrative monitoring.
- Analytics reports and dashboard data are prepared for system insights.

Actors –

- Administrator
- Database

Flow of Events

1. Basic Flow

1. The administrator accesses the system login page.
2. The administrator enters login credentials.
3. The system authenticates the administrator.
4. The system displays the administrative dashboard.
5. The administrator selects a management function.
6. The administrator may manage course information (add, update, or delete courses).
7. The administrator may manage professor records and course assignments.
8. The administrator may monitor feedback reports generated from student evaluations.
9. The system generates analytics reports based on stored feedback data.
10. The system prepares dashboard analytics data for visualization.
11. The administrator views the generated reports and analytics information.

2. Alternative Flow

1. The administrator may directly generate analytics reports without modifying course or professor data.
2. The administrator may choose to only view feedback reports.

3. Exceptions

1. Invalid administrator login credentials prevent access to the system.
2. Database connectivity failure prevents retrieval or modification of system data.
3. Missing feedback data results in empty analytics reports.

Includes –

- Analytics Reports
- Generate Analytics Reports

Notes/Issues –

- Administrative access must be restricted to authorized users only.
- All modifications to course and professor records should be logged to maintain system accountability.

Other Non-functional Requirements

3.4 Performance Requirements

The system must provide efficient performance to ensure smooth interaction for all users accessing the Professor Rating Based on Course Outcome Feedback System.

P1. Response Time - The system shall respond to user requests such as page loading, login, and course retrieval within 3 seconds under normal operating conditions.

P2. Concurrent User Handling - The system shall support at least 200 concurrent students accessing the platform simultaneously during peak feedback submission periods.

P3. Feedback Submission Processing - The system shall process and store submitted feedback within 2 seconds after the user submits the feedback form.

P4. Data Retrieval Performance - The system shall retrieve and display course lists, feedback summaries, and reports efficiently without noticeable delays or stutters.

3.5 Safety and Security Requirements

The system must ensure that user data, feedback information, and system operations remain secure and protected from unauthorized access.

S1. User Authentication - The system shall require users to authenticate using valid institutional credentials before accessing system features.

S2. Feedback Anonymity Protection - The system shall ensure that feedback submitted by students remains anonymous and cannot be traced back to individual users.

S3. Data Protection - The system shall securely store all system data including student records, course information, and feedback data a secure database where only authorized users can access the data.

S4. Access Control - The system shall enforce role-based access control to ensure that students, professors, and administrators can only access functionalities relevant to their roles.

S5. Secure Communication - The system shall support secure communication between the client interface and backend services using standard web security protocols.

3.6 Software Quality Attributes

1.Reliability - The system shall ensure reliable operation so that users can submit feedback without system failures or data loss. Feedback data shall be stored safely in the database to prevent accidental loss of submitted responses.

2.Usability - The system shall provide a simple and intuitive user interface that allows students, professors, and administrators to easily navigate the platform and perform their tasks with minimal training.

3.Maintainability - The system shall be designed using a modular architecture separating the frontend, backend services, and database components to simplify future maintenance and updates.

4.Scalability - The system shall support future expansion to accommodate increasing numbers of users, courses, and feedback records without major architectural changes.

4 Other Requirements

1. Database Requirements

The system shall use MongoDB as the primary database for storing system data. The database shall store information including student records, course details, professor information, attendance data, and submitted feedback. The database structure shall support efficient retrieval of feedback data for reporting and analysis.

2. Data Backup and Logging

The system shall maintain logs of important system events such as user login attempts, feedback submissions, and administrative actions. These logs help in monitoring system activity and identifying potential system issues. Periodic database backups shall be performed to prevent loss of feedback data.

Appendix A - Data Dictionary

Data Element	Description	Type	Related Operations	Related Requirement
Student_ID	Unique identifier assigned to each student in the university system	String	Used during authentication and course retrieval	F1, F2
Student_Name	Name of the student enrolled in the university	String	Used for display in admin records (not linked with feedback)	F2
Course_ID	Unique identifier for each course offered by the university	String	Used to retrieve course information and feedback forms	F2, F4
Course_Name	Name of the course associated with a professor	String	Displayed in student dashboard and feedback forms	F2
Professor_ID	Unique identifier for professors	String	Used to associate feedback with professors	F5, F12
Professor_Name	Name of the professor teaching a course	String	Displayed in course feedback form and reports	F5
Attendance_Percentage	Percentage of student attendance for a specific course	Float	Used to verify feedback eligibility (≥75%)	F3
CO1 – CO5 Ratings	Ratings provided by students for each Course Outcome	Integer (1–5 scale)	Used to evaluate how well course outcomes were delivered	F6
Feedback_Comment	Optional textual comment submitted by a student	Text	Stored anonymously in the database	F7
Feedback_ID	Unique identifier for each feedback submission	String	Used for storing and retrieving feedback data	F8
Submission_Timestamp	Date and time when feedback is submitted	Date Time	Used for system logs and auditing	F10
Admin_ID	Unique identifier for administrators	String	Used for admin authentication and logging	F14
Feedback_Report	Aggregated statistical summary of feedback for a course	Data Object	Generated for professors and administrators	F12, F16

Appendix B - Group Log

Date	Participants	Activity	Outcome
9/3/2026	All Group Members	Initial discussion of project idea and scope	Decided to build a Professor Rating System based on Course Outcome feedback
10/3/2026	Rishith, Sreeja, Srujana	Discussed SRS structure and what to mention in each section	Finalized document structure and task distribution
11/3/2026	Sreeja, Srujana	Drafted Introduction and Overall Description sections	Completed Draft 1 of SRS
12/3/2026	Rishith, Vijay	Worked on Functional Requirements and System Description	Defined system functionalities and requirements
13/3/2026	Rishith, Vijay	Added Non-functional requirements and constraints	Completed Draft 2 of the SRS
15/3/2026	All Group Members	Discussion of system architecture and UML diagrams	Finalized use case diagrams and system components
16/3/2026	All Group Members	Document review and finalization	Completed Draft 3 and prepared final SRS submission